Speech-Eval

Contrastive Speech Delivery Analytics with Temporal Flaw Grounding

Multimodal AI Hackathon 2026 - Track C | October 2026

1. Problem and approach

Speech-Eval shows you delivery problems such as speaking too fast, too quietly, or pausing in an awkward place. It marks the affected words and times so you can listen to the exact part. It explains each flag using measured differences from reference readers or your earlier take.

Frozen A: precision 0.125, recall 0.400, F1 0.190.
Frozen B: precision 0.018, recall 0.390, F1 0.035.
A clean alarms: 1.341/min; dev guard failed (1.777 versus <=0.5/min).
Release: 1,103 indexed recordings, 36.3 h; repeated scientific hashes match.
Limits: low precision; genuine labels are recorder-confirmed, not independent; zero rater responses.

Content quality is out of scope. Controlled edits supply reproducible operation labels, not independent judgments of good speaking. A uses same-text train readers; B uses the speaker's earlier take; C-lite uses sentence norms without a same-text reference. Key-moment messages are labelled heuristics, not scored judgments.

2. Contrastive dataset construction

Sources: 15 Gettysburg and 13 O Captain LibriVox readings, plus one JFK Rice and one Reagan Challenger recording. USA public-domain evidence is recorded in the release manifests and dataset card; automatic evidence checks are not human legal review. Raw hashes remain immutable.

Canonical/as-spoken differences were checked with local CTC candidate likelihoods calibrated on 300 controls, empirical fake acceptance ≤2.0%. This in-sample proxy is not listening or guaranteed accuracy on real disagreements. Unsettled words and dependent evidence remain masked. Number expansion retains index mappings. Acoustic audit.

MFA/WhisperX were compared on 37 words / 74 edges after 23 unclear-word exclusions. Boundaries were placed visually from waveform and spectrogram by an AI assistant (Claude), without listening; a human spot-check is pending. Independence from model templates was asserted, not independently established. Pooled median differences: MFA 52.0 ms, WhisperX 72.0 ms; MFA is primary. Both passed independent CPU rerun identity. These are differences from a limited visual reference, not established human accuracy. Bake-off.

Eligible sources receive seven effects at five edit strengths plus mixed variants, with seeded placement. One clipped O Captain source is audit-only. The release has 1,103 indexed variants / 36.3 h: 30 originals and 1,073 edited recordings. Raw downloads/duplicate rates are not counted again. Train/dev/test have 18 / 4 / 7 speakers. Measured counts.

Type	Controlled operation	Increasing edit strength
rushing / dragging	Pitch-preserving stretch	Raise / lower speed factor
monotone	PSOLA around median pitch	Retain less pitch range
too_quiet	Flat/trailing gain	More attenuation

Type	Controlled operation	Increasing edit strength
awkward_pause	Phrase-internal room tone	Longer inserted gap
missing_pause	Shorten existing gap	Lower retained fraction
mumbling	Zero-phase low-pass plus gain	Lower cutoff/gain

Parameters are in configs/flaws.yaml. Time maps transport source indices into the flawed timeline; labels bind type, edit-strength severity and interval. Real MFA is run separately for detection. Oracle alignments serve label/QA only; edit boundaries are not guaranteed perceptual boundaries. Genuine recorder-confirmed takes are evaluated separately below, without threshold tuning.

3. Features and speaker normalisation

Praat pitch/intensity/HNR and spectral/MFCC tracks feed word, short-window and sentence aggregates. Speaker-relative transformations reduce voice/microphone offsets, not all differences:

$st(t) = 12 \log_2[f_0(t) / \text{median } f_0]$; $Lrel(t) = L(t) - \text{median } L$.

Medians use voiced frames inside aligned words. Rate (syllables/second) is not speaker-normalised. Clarity is a high-band energy ratio, not intelligibility. Comparison rate/pitch-variation/pause features are log-transformed; loudness and clarity remain dB. Uncertain-word masks exclude dependent aggregates. Feature normalisation does not silently edit stored loudness.

4. Detection, grounding and explanations

Clean train readers give word/feature median mu and robust spread s = 1.4826 median(|x - mu|). Pooled train spread and shrinkage n0 = 5 give:

$\sigma^2_i,f = (n\, s^2_i,f + n_0\, \sigma^2 pool,f) / (n + n_0)$; $z_j,f = (x_j,f - \mu_i,f) / \sigma_i,f$.

Whole-word mapping excludes unmatched/partial correspondences. Runs within sentences become regions; pauses are boundary events. Directional feature rules and contradictory-pair suppression are explicit. Frozen selected k is **3.50 for every type**, not the draft default 2.0. Dev severity cut points are [2.0, 2.75, 3.5, 4.059, 6591.0]; matched dev Spearman is 0.16. The unusual final cut point/weak correlation are retained, not repaired with test or user takes. Calibration.

Templates quote measured participant/reference deviations and flagged words. The following is **an example of real system output**, quoted verbatim from the presentation layer using the unchanged numeric evidence in analysis 86964ad39ead, Mode A test file gettysburg__lv_sa__awkward_pause_s3_01:

> 75.99–77.58 s, "it can": you paused 1.59 s between "it" and "can", where good readers do not pause (1.59 s longer, more than 10 SD above). It breaks the sentence.

The detected interval is 75.990-77.580 s, participant pause 1.590 s, reference shown as no pause (less than 0.02 s), deviation shown as more than 10 SD. Full-precision reference/z remain in the JSON. Its matching injected label spans 76.035-77.535 s in the label file. It is a correct hit under the frozen pause-centre rule. The extreme z reflects near-zero reference spread; it is not evidence of that much perceptual severity. This existing test example is for report illustration, not v2 tuning. Measured rule explanations are **not causal proof**.

B uses the own earlier take; JFK/Reagan test originals are within-speaker references, not population training/calibration. C checks sentence rate, pitch range, final loudness drop and non-final end pause; mumbling/awkward pause are unsupported. Number/end guidelines never alter scores.

5. Scoring and validation

For first-to-last-word duration D, duration penalty is sum(v × d/D × 10); pause penalty is sum(0.4 × v). Clip 10 - penalty to [0,10], decimal half-up round, then weight rounded categories: pace/pitch 0.25 each, pauses 0.20, volume/clarity 0.15 each. These are configurable design choices, not an external standard. Severity-3 rushing over 10.0% gives pace 7.0. Validation.

The report gives per-type, not aggregate, monotonicity:

Type	Median severity-score Spearman	Non-increasing %	Constant series
awkward_pause	-0.87	85.7	3
dragging	-0.71	71.4	2
missing_pause	-0.71	100.0	4
monotone	-0.71	28.6	0
mumbling	-0.89	71.4	0
rushing	-0.71	71.4	1
too_quiet	-0.71	85.7	3

Each type has seven complete series. Constant ranks have undefined Spearman; non-increasing alone is not sensitivity. Human agreement is pending: zero independent responses. The blind pack/scripts are ready; AI and recorder self-ratings are not used as independent rater data. The form asks overall delivery 1–5, flaw type(s) and approximate time, not five category ratings. Scripts compute system–mean-human/per-rater Spearman, mean defined pairwise human Spearman and exact flaw-set agreement with denominators; no fabricated responses or CIs.

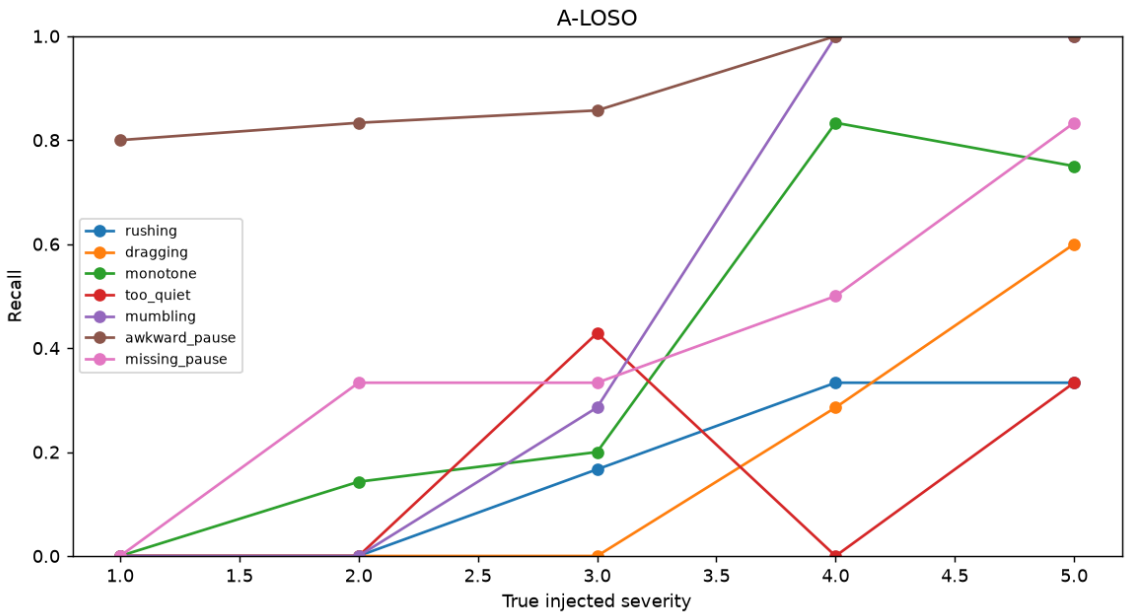
Frozen host two-run identity: 10/10. Later fresh-clone CPU Docker reproductions matched all 563 scientific outputs twice, including cross-environment checks. Scope and SHA manifests.

6. Results across settings

Thresholds used dev only. Original final test commit: aaf1832fe24a193d4c90f9e7a2ff9537b753335c. Later test runs are **reproductions**, not tuning. Same-type one-to-one region IoU ≥0.3; pause centre distance ≤0.25 s. Frozen results.

Setting	Files	Precision	Recall	F1 [95% CI]	Clean alarms/min
A-LOSO	190	0.125	0.400	0.190 [0.156, 0.223]	1.341
B-same-speaker	76	0.018	0.390	0.035 [0.026, 0.045]	0.000

A highest recall: awkward_pause 0.903; lowest: rushing 0.167. Matched overall onset/offset differences are 26.0 / 40.5 ms (A), 31.0 / 35.5 ms (B), conditional on matches, not all truth regions. Dev clean-alarm guard **failed**: 1.777 versus ≤0.5. Learned detector was not trained. No pooled severity-1/2 or severity-4/5 recall is inferred from the chart without a reported numeric summary.



A-LOSO recall versus injected severity

6.1 Secondary detector v2

post-hoc v2; test split was seen with v1, so results are optimistic. V1 above remains the primary frozen result. V2 was selected from exactly 30 DEV configurations and frozen at 95c8f334faf4c134fcbf5b4fa31fda4635f336ea before one marker-protected test evaluation. No detector/configuration changes followed. Within-recording sentence contrast, practical effect minima and region constraints raised DEV F1 from 0.150 to 0.219 with 0.410 clean alarms/min, passing the declared promotion gate. The selected configuration SHA is 3496a9c3....

Setting / version	Files	Precision	Recall	F1 [95% CI]	Clean/min
A frozen v1	190	0.125	0.400	0.190 [0.156, 0.223]	1.341
A post-hoc v2	190	0.145	0.288	0.193 [0.154, 0.232]	0.671
B frozen v1	76	0.018	0.390	0.035 [0.026, 0.045]	0.000
B post-hoc v2	76	0.042	0.207	0.070 [0.043, 0.099]	0.000

Rows, same-type IoU/pause-centre matcher, uncertainty exclusions and denominators are identical; no uncertain truth regions were excluded on these test rows. Full per-type and severity comparison reports precision/recall/F1, boundary differences, clean alarms and severity Spearman for both detectors. Strength-cohort precision/F1 uses shared clean controls and homogeneous eligible truth strengths; mixed-strength files are excluded from those cohorts but included in full-set recall by severity. Full precision and frozen hashes remain in v2 JSON. For A, v2 halves clean alarms (1.341 to 0.671 per min) but lowers recall (0.400 to 0.288). F1 is effectively unchanged (0.190 to 0.193); v1 stays primary. The test clean-alarm guard still fails. B precision/F1 rise but remain low, with recall falling. V2 is a precision/recall trade-off, not established general improvement; uniform recording-wide problems can be missed by within-recording contrast. Human takes/ratings were not used for tuning.

7. No-reference comparison

C uses **different eligibility and sentence/type matching**. Its paired comparison on 175 files stays separate from Phase6:

Paired setting	Precision	Recall	Clean alarms/min
Same-text / own-original reference	0.106	0.324	0.584
C-lite norms	0.020	0.179	4.159

C too-quiet truth is trailing-only; unsupported single effects are omitted and mixed files score supported types only. Lower aggregate recall/more clean alarms describe controlled edits, not humans. Mode C.

7.1 Genuine recorder-confirmed takes

Eight consented takes completed real MFA: A for all, B02/03 versus own01, C for both spontaneous take04s (unknown flaws, excluded from scoring). Labels are recorder-confirmed, not independent; word anchors were fixed before results. Transcripts apply ASR structural edits but keep canonical substitutions and mumbling words; verbatim fidelity is not independently verified. No tuning.

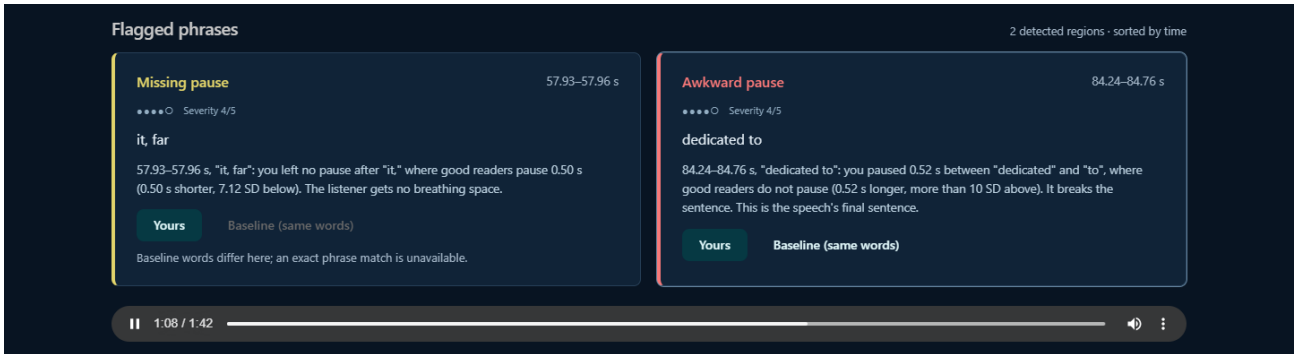
Version / mode	Labels	Hits / misses / FP	FP/min
Frozen v1 / A	13	4 / 9 / 77	9.625
Frozen v1 / B	13	3 / 10 / 60	11.236
Post-hoc v2 / A	13	4 / 9 / 58	7.250
Post-hoc v2 / B	13	3 / 10 / 27	5.056

Primary matching is same-type one-to-one IoU >=0.3, including pause events. A pools six scored takes (8.00 min); B pools four current takes (5.34 min), not their references. Take01 flags: Gettysburg/Reagan 9/14 (v1), 7/11 (v2). Historical pause-centre <=0.25 s, secondary unit overlap, MFA spans, transcripts and dashboard links: Phase10. V1 stays primary; post-hoc v2; test split was seen with v1, so results are optimistic. Limited hits/high false alarms do not establish independent perceptual accuracy.

Takes finish sooner than every same-text reference: Gettysburg 80.99-86.81 s versus 102.88-190.95 s; Reagan 76.42-80.29 s versus a 90.17 s excerpt. Total aligned gap seconds are lower than nine Gettysburg readers and Reagan; only Gettysburg take03 is below the tenth reader too. This is not every sentence or pause. Mode A's 53 unlabelled missing pauses (of 77 v1 false alarms) are measured reader differences, counted as false alarms because unlabelled, never hits.

8. Dashboard

The results view combines the waveform, flagged spans, rubric scores and generated explanations. Selecting a region isolates its words and supports participant and same-words baseline playback. Screenshots are from the verified real dashboard, not new performance measurements. Browser checks.



Selected flaw region and same-words playback controls (excerpt; full screenshot)



Dashboard results: waveform and score card (excerpt; full screenshot)

9. Limitations and future work

Controlled edits are not real co-occurring flaws, and neutral reading is not ideal oratory. Few texts/speakers and English-only proxies limit generalisation. The unheard visual reference and unclear-word exclusions introduce uncertainty and selection bias; model-to-visual differences do not establish annotation accuracy. C/key-moment heuristics and rubric scores lack human validation. Low precision, the failed dev guard, constant/non-monotonic series and fixture miss remain limits. Future work needs more consented recordings and independent labels/ratings; detector changes must be labelled post-hoc, not presented as frozen evaluation.

10. Reproducibility and attributions

make reproduce-quick DATA_ARCHIVE=/absolute/path/speech-eval-dataset-v1.zip imports the separately supplied archive and runs one text/three test readers. make reproduce re-verifies inputs and reproduces frozen Phase6/7/8 numbers in pinned CPU Docker, not every historical ASR/excerpt decision or human/video work. Quick: 11.829 / 11.954 seconds; full: 225.204 / 223.142 seconds after import/build. Hashes: data/release/reproduction_hashes.json and reproduction report. The narrow legacy-fixture gate exemption is a disclosed post-hoc release adapter, not a detector change; other historical inputs/code are verified.

Audio attribution: LibriVox volunteers and JFK/Reagan institutional collections; source-specific USA public-domain evidence. Full-precision scientific values remain in the original JSON; display rounding does not change them.

Pending items

Independent agreement pending (zero responses); human alignment spot-check.

Archive upload URL, author confirmation and recorded/published video URL.

Owner-confirmed formal code/annotation licence files; total disk requirement.

Unreported aggregate severity/C-F1 and pitch-improvement values remain unclaimed.

See NEXT and placeholder inventory.